Hybrid Movie Recommender

Cold-start handling, one-line reasons and a discovery dial, built and tested on MovieLens



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The problem

Cold start

A new user has no history, so most recommenders fall back to the same generic list.

No reasons

People get a list of titles and no idea why each one was picked, so they stop trusting it.

Endless scrolling

Catalogs are huge. People spend more time scrolling than watching.

Goal: useful picks within a minute of signing up, each with a reason the user can check.

Six signals, one score

Preference

Genres from the quiz

Content

TF-IDF on genres and tags

Collaborative

Films liked by similar users

Context

Tonight's mood

Quality

Rating and popularity, capped

Discovery

Unfamiliar but fitting

Weights depend on how much the user has interacted: cold (nothing yet), warming (1 to 19 likes or dislikes), established (20 or more). Each signal is min-max scaled inside the candidate set before weighting.

An MMR re-rank then sets how varied the row is, and the top two signals write the reason shown on each card.

Data and test setup

9,125	100,004	671	1,296
movies	ratings	users	tag applications

MovieLens ml-latest-small (2016 release), GroupLens Research

Time split per user	First 80% of each user's ratings train the models. The last 20% are held out as the test.
What counts as a hit	A held-out film rated 4 or higher. Films the user already rated in training are never recommended.
Baselines	Popularity only, content only, collaborative only and our own hybrid with the spec weights.

From signals to a ranked row

1

Candidates

Every film with 5 or more ratings that the user has not seen

2

Hard filters

Disliked, seen and avoided-genre films are removed

3

Score

Six signals times the stage weights, minus a repeat penalty

4

MMR re-rank

Score minus similarity to films already in the row; no genre above 50%

5

Explain

The top two signals fill a reason template

Discovery dial: Comfort zone uses MMR lambda 0.85, A little different 0.70, Surprise me 0.55, and raises the discovery weight from 0.05 to 0.10 to 0.20.

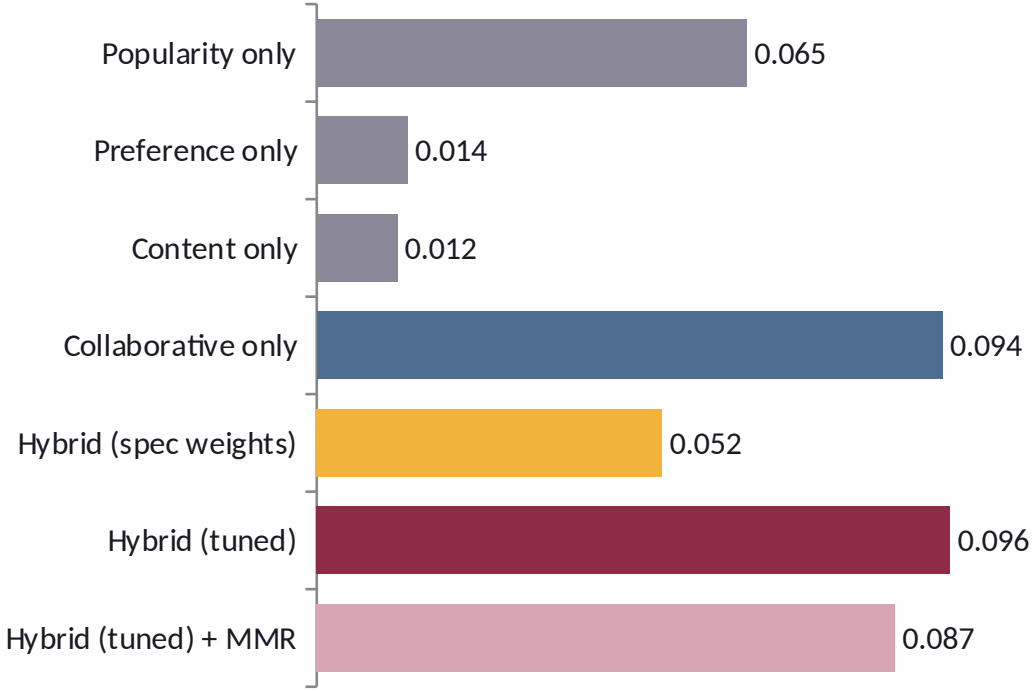
Weights by profile stage

Signal	Cold	Warming (spec)	Warming (tuned)	Established (spec)	Established (tuned)
Preference	.45	.30	**.13**	.20	**.098**
Content	.15	.25	**.057**	.25	**.065**
Collaborative	.00	.10	**.424**	.25	**.448**
Context	.15	.15	.15	.15	.15
Quality	.15	.10	**.139**	.05	**.139**
Discovery	.10	.10	.10	.10	.10

The spec weights were starting values. We tuned the warming and established columns by random search on a validation slice cut from the training data, never the test set. Quality is capped at 0.15 and every signal keeps at least 0.05. The cold stage has no data to tune on, so it keeps the spec values.

Results: users with history

NDCG@10 on the held-out last 20% (585 users)



+47%

NDCG@10 vs popularity (paired t-test $p < 0.001$). Recall@10 +44%

+1%

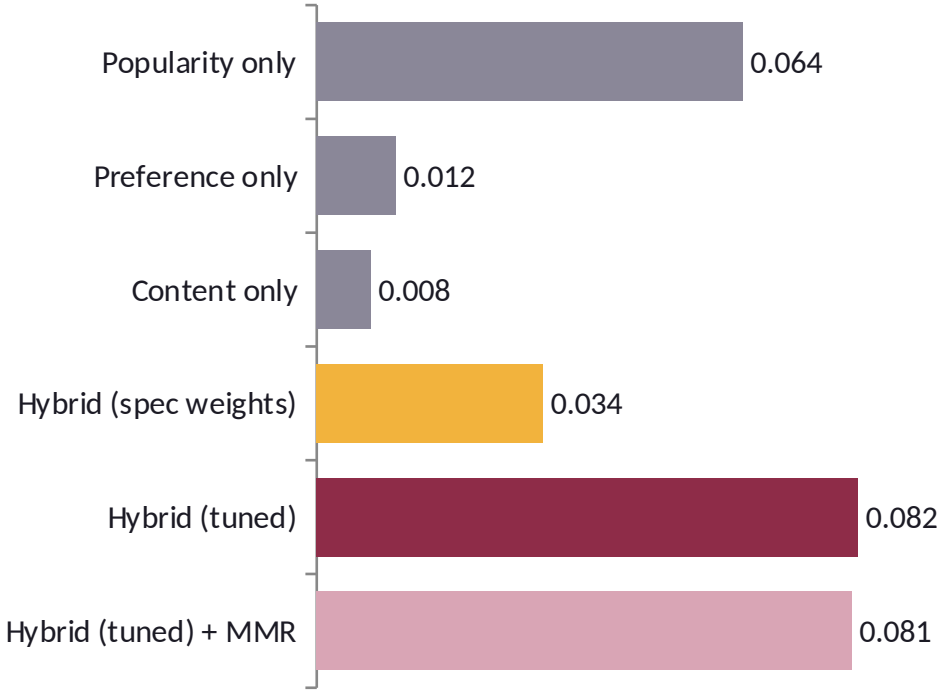
vs collaborative only. No real difference ($p = 0.63$)

0.052

NDCG@10 with the untuned spec weights, below popularity

Results: cold start

NDCG@10 using only 5 favourites and their top 3 genres (581 users)



+27%

NDCG@10 vs popularity (p = 0.002). Our spec target was +15%

10x

the content-only baseline, which has little to work with from 5 films

5 films

is all the model knows about each user here

Accuracy against variety

Method (full history)	NDCG@10	Genre diversity	Catalog coverage
Popularity only	0.065	0.796	4%
Content only	0.012	0.194	30%
Hybrid (tuned)	**0.096**	**0.724**	**13%**
Hybrid (tuned) + MMR	**0.087**	**0.827**	**13%**

+0.10 diversity

MMR costs 9% of NDCG@10 (p = 0.001). The dial lets the user choose that trade.

13% coverage

Below the 20% target in our spec. Collaborative scores concentrate on well-rated films, which probably explains it.

Live example: five sci-fi favourites

Comfort zone picks	Reason shown on the card
Star Wars: Episode V - The Empire Strikes Back (1980)	Popular with people who liked Matrix, The (1999). Rated 4.2/5 by 234 viewers.
Star Wars: Episode IV - A New Hope (1977)	Popular with people who liked Matrix, The (1999). Rated 4.2/5 by 291 viewers.
Usual Suspects, The (1995)	Popular with people who liked Blade Runner (1982). Rated 4.4/5 by 201 viewers.
Alien (1979)	Popular with people who liked Terminator, The (1984). Rated 4.0/5 by 127 viewers.
Pulp Fiction (1994)	Popular with people who liked Blade Runner (1982). Rated 4.3/5 by 324 viewers.

Same user, Surprise me

Keeps 3 of 5 and swaps in:

- Schindler's List (1993)
- Shawshank Redemption, The (1994)

Same five favourites. Only the dial changed.

Run live: streamlit run app.py. Click Like or Not for me and the row re-ranks at once.

Limits and next steps

Limits

- Catalog coverage is 13%, below our 20% target
- On accuracy the hybrid ties collaborative only; the clear win is over popularity and content
- Small, older MovieLens release; weights tuned on one split
- No runtime or language filters, because the data has neither

Next steps

- Full stack from the spec: Next.js, FastAPI and Postgres
- Train on MovieLens 25M and add a franchise cap
- Group movie night and free-text requests
- Track pick rate and negative feedback with real users

Questions

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Code, tests and evaluation: python eval.py, python -m pytest, streamlit run app.py

